amazon rme

AMAZON BASE BUILDING MAINTENANCE STANDARD

GLOBAL CENTRAL RME STANDARDS & COMPLIANCE

Task Title	ELECTRICAL SYSTEM, DIESEL POWER GENERATORS - 11
APM Object	ELEC.GEN
APM EC	AMZ.ELEC.GEN
BBM Handbook Title	Diesel Power Generators
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure These tasks are based on industry guidance, SME knowledge and HIE learning. Please ensure vendor specific requirements are followed. To suggest improvements please raise a SIM SIM - Link to SIM template

Last Update: (Version)	Jul-23	**Geo Applied:**	Intl APM Instance	**F-Gas**	Not Identified	**Activies may be covered by No-Go restrictions:**	TBC	**Special Certificaion Required:**	TBC	**Delivered By:**	Carried out by trained on site technician certified in the electrical field	**OCB:**	BBM
Evidence for CT:	YES	**Pricing template Position:**	11	**Related CT Areas:**	CT.ELEC.POWERGENERATOR			**Assets included in this PM:**	DIESEL GENERATOR UNIT	**Schedule With:**	-		

Description of System / Technology (According to BBM Handbook)

A diesel generator is the combination of a diesel engine with an electric generator (often an alternator) to generate electrical energy.
A temporary generator connection box provides a safe and efficient means of connecting a commercial or industrial business with a portable power source in the event of an electrical outage. A generator connection box is most frequently situated on the exterior of a building, wall- or floor-mounted, and used in conjunction with a transfer switch or kirk keyed breaker on switchboard.

Additional Information

All sites with the above listed Base Building Asset should conduct routine maintenance and inspections according to this standard PM package. The Amazon maintenance standard supports compliance, reduces safety risks,improves asset reliability & efficiency.
Some maintenances captures several assets within one PM package -> Ensure that corrective/reactive maintenance for assets where proactive maintenance is contained under another asset is created against the original asset.

Task Index	Component	Tasks	Tasks Nested for frequencies > weekly									
			Daily	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	36 Monthly	48 Monthly	60 Monthly
1	General check	Is the room properly ventilated						PM				
2	General check	Are the components clean - free from debris & leakage						PM				
3	General check	Check of the equipment for damage & missing components						PM				
4	General check	Check oil tanks for damage & leakages						PM				
5	General check	Check collection tray for deposits -> identify the root cause in case of deposit						PM				
6	General check	Are there any alarms indicated						PM				
7	General check	Did any circuit breaker trip						PM				
8	Documentation	Are the circuit diagrams and relevant technical documentation available and complete						PM				
9	Documentation	Are there manual entries in the drawings or technical documentation that indicate changes to the installation						PM				
10	Safety	Are all relevant safety notices visible and free from damage						PM				
11	Safety	Is appropriate fire-fighting equipment available in the rooms						PM				
12	Safety	Are disconnecting devices lockable in the open position (LOTO)						PM				
13	Electrical components	Visual check of cable connections such as pinched insulation in the contact area						PM				
14	Electrical components	Visual check for the protection against direct contact with live parts						PM				
15	Electrical components	Were any obvious changes made (e.g. rewiring, hanging cables)						PM				
16	Electrical components	Are all cables properly routed						PM				
17	Electrical components	Are all components properly labeled or marked						PM				
18	Electrical components	Test all safety relevant switches						PM				
19	Electrical components	Are all components properly attached						PM				
20	Electrical components	Ensure all cable connections are correctly torqued according to manufacturers specification						PM				
21	Electrical components	Are all metal parts connected to the protection earth (ground)						PM				
22	Electrical components	Is the ground connection to the building ground as designed?						PM				
23	Battery & Charger	Check if the battery charging device is working correctly & is free from damage						PM				
24	Battery & Charger	Measure the voltage of each battery (disconnected) and plan for replacement if necessary						PM				
25	Battery & Charger	Check the batteries for leakages						PM				
26	Battery & Charger	Check the fluid level of the batteries and recharge if required						PM				
27	Battery & Charger	Schedule a battery change for any kind of deviation from standard or design						PM				
28	Fuel supply	Check the fuel tank and supply pipes for leakages and damage						PM				
29	Fuel supply	Check fuel filter and change if required						PM				
30	Fuel supply	Check all valves including safety valves for correct function						PM				
31	Engine cooling system	Is the level of the cooling fluid ok -> refill if required						PM				
32	Engine cooling system	Check for proper corrosion and freezing protection of the cooling fluid						PM				
33	Engine cooling system	Check hoses & pipes for damage & wear						PM				
34	Engine cooling system	Check connections at hoses & pipes for tightness						PM				
35	Engine cooling system	Check cooling water pre heater for correct function						PM				

36	Engine function	Exchange oil & oil filter						PM					
37	Engine function	Check valve setting and readjust if necessary						PM					
38	Engine function	Check vibration decoupling for damage						PM					
39	Engine function	Check V-belt tension and wear						PM					
40	Engine function	Check the function of the limit & safety switches						PM					
41	Engine function	Check injection system for damage and correct function						PM					
42	Engine function	Check correct settings of the speed controller						PM					
43	Engine function	Check the engine starter and alternator for correct function						PM					
44	Engine function	Check all value transmitters and display elements for damage and correct function						PM					
45	Engine function	Check Engine clutch for function						PM					
46	Engine function	Are there any unusual noises during start, operation or stop of the engine						PM					
47	Engine function	Grease all possible greasing points						PM					
48	Engine function	Check that the screws and nuts are tight						PM					
49	Engine function	Check engine block for leakages						PM					
50	Engine function	Is the exhaust tight and free from damage						PM					
51	Electrical function	Check the automatic & correct function of all air in- & outlet flaps			PM			PM					
52	Electrical function	Check the synchronization & automatic transfer switch for correct function (Test with load, only if there won`t be any impact on connected assets			PM			PM					
53	Electrical function	Functional test with highest possible load (Test with load, only if there won`t be any impact on connected assets			PM			PM					
54	Electrical function	Is the generator voltage and current flow as designed/demanded			PM			PM					
54	Electrical function	Check emergency shut off system for correct function						PM					
55	Operational Hours	Record the operational hours of the unit into the notes			PM			PM					
56	Functional cleaning	Clean the air intakes on the engine, generator and main air intake grilles on the outer walls						PM					
57	Functional cleaning	Carry out a functional cleaning of all equipment if required						PM					
58	Final Check	Check the O&M manual and carry out additional required tasks						PM					

Change Log

July 2023 - First issue in new format. Simplification of task scope, alignment with CT.
Aug 2024 - Added task to capture the operating hours of the unit for Sustainability purposes